

BUSINESS INSIGHTS & RECOMMENDATIONS

HealthConnect Experience Lab — Week 5 | Appointment Attendance Analytics

Professional English Version | Data Analytics Track

EXECUTIVE DECISION SUMMARY

The analysis identifies a high overall exposure to no-shows: 51.15% of non-cancelled appointments result in a no-show. Among the factors examined, booking lead time shows the clearest observed differentiation in No-Show Rates. Recorded previous no-shows provide a useful supporting history signal. Reminder exposure shows limited differentiation, while several temporal and demographic characteristics do not currently justify standalone prioritisation.

1. BUSINESS INSIGHTS — EVIDENCE, INTERPRETATION & IMPLICATIONS

01 — HIGH OVERALL NO-SHOW EXPOSURE

Finding: There are 2,423 no-shows among 4,737 non-cancelled appointments, resulting in an overall No-Show Rate of 51.15%. The 263 cancelled appointments are excluded from the denominator of this KPI.

Interpretation: No-show is the most frequent appointment outcome within the analytical population used for attendance analysis.

Business implication: Reducing missed appointments warrants operational attention, but interventions should be targeted using factors that meaningfully differentiate No-Show Rates rather than applied uniformly.

02 — BOOKING LEAD TIME SHOWS THE CLEAREST OBSERVED DIFFERENTIATION

Finding: The No-Show Rate rises from 29.47% for appointments booked 0–7 days in advance to 71.36% for appointments booked 46–60 days in advance. The 31–45 day and 46–60 day bands record rates of 57.03% and 71.36%, respectively, compared with the overall 51.15% reference.

Interpretation: Observed no-show exposure increases materially as the time between booking and appointment lengthens.

Continuous analysis also shows a positive monotonic association (Spearman $\rho = 0.2873$).

Business implication: Longer booking lead times represent the first scheduling context to investigate for targeted pre-appointment engagement and attendance-support actions.

03 — RECORDED PREVIOUS NO-SHOWS PROVIDE A USEFUL SUPPORTING HISTORY SIGNAL

Finding: The No-Show Rate increases from 46.30% when previous_no_shows = 0 to 69.74% when previous_no_shows = 3, producing a calculated gap of +23.43 percentage points. The overall relationship is positive but limited in strength (Spearman $\rho = 0.1214$).

Interpretation: Recorded previous no-shows differentiate appointment outcomes more clearly than the simple number of previous appointments, while remaining a supporting signal rather than a standalone predictor.

Business implication: This recorded history can strengthen prioritisation logic, particularly when combined with scheduling contexts that already show higher observed no-show exposure.

04 — REMINDER EXPOSURE SHOWS LIMITED DIFFERENTIATION, NOT EVIDENCE OF EFFECTIVENESS

Finding: Reminder Coverage Rate is 72.68%. Among non-cancelled appointments, the No-Show Rate is 54.63% without a recorded reminder versus 49.86% with a reminder, a descriptive gap of 4.78 percentage points. The association is statistically detectable but very weak (Cramér's $V \approx 0.042$).

Interpretation: The data show a difference by reminder exposure, but they do not establish that reminders caused the observed difference.

Business implication: Reminder strategy should be investigated further — including targeting, timing and channel — before drawing conclusions about effectiveness or changing operational policy.

05 — SEVERAL FACTORS DO NOT SUPPORT STANDALONE PRIORITISATION

Finding: Appointment day, appointment time, estimated waiting time, age group and gender do not show statistically or decisionally meaningful differentiation in No-Show Rates. Distance to clinic shows only a weak association and requires cautious interpretation.

Interpretation: The available factors contribute unevenly to understanding no-show patterns; not every measurable difference is decision-relevant.

Business implication: Avoid standalone targeting or intervention rules based solely on these variables. Analytical and operational attention should first focus on signals supported by stronger and more stable evidence.

2. RECOMMENDATIONS — FROM EVIDENCE TO ACTION**01 — PRIORITISE LONG-LEAD APPOINTMENTS**

WHAT	Test enhanced pre-appointment engagement for appointments scheduled well in advance.
WHERE	Prioritise the 31–45 day band and, especially, the 46–60 day booking lead band.
WHY	Both groups exceed the overall No-Show Rate of 51.15%, with 71.36% observed for appointments booked 46–60 days in advance.
HOW TO MEASURE	Track No-Show Rate by Booking Lead Band and compare the evolution of prioritised groups with their observed baseline.
NEXT STEP	Define a limited operational test before wider implementation, specifying contact approach, timing and eligible appointment population.

02 — USE RECORDED PREVIOUS NO-SHOWS AS A SUPPORTING PRIORITISATION SIGNAL

WHAT	Incorporate recorded previous no-shows as a complementary prioritisation signal, not as a standalone rule.
WHERE	Appointments with a recorded no-show history, particularly when combined with longer booking lead times.
WHY	The No-Show Rate reaches 69.74% for three recorded previous no-shows versus 46.30% for zero.
HOW TO MEASURE	Track No-Show Rate by recorded previous no-shows and, in a subsequent phase, evaluate combinations with booking lead time.
NEXT STEP	In Week 6, assess whether combining these signals improves prioritisation while avoiding over-interpretation of sparse higher-history categories.

03 — EVALUATE REMINDER STRATEGY BEFORE MODIFYING IT

WHAT	Investigate reminder targeting, timing and channel rather than assuming reminder effectiveness from the observed association.
WHERE	Compare appointments with and without reminders, then compare SMS, WhatsApp and Email among reminded appointments.
WHY	The observed reminder-status gap is 4.78 pp, while the association strength remains very weak.
HOW TO MEASURE	Retain Reminder Coverage Rate and No-Show Rate by Reminder Status; add more controlled comparisons if future data permit.
NEXT STEP	Document current reminder-allocation rules and identify additional variables required for a more robust effectiveness assessment.

04 — AVOID TARGETING BASED ON WEAKLY DIFFERENTIATING FACTORS

WHAT	Do not build standalone no-show reduction strategies around appointment day, appointment time, estimated waiting time, age group or gender based on the current evidence.
WHERE	Segmentation design, prioritisation rules and future analytical prototypes.
WHY	These variables did not show sufficiently robust differentiation to justify independent targeted action.
HOW TO MEASURE	Retain them as contextual or control variables and reassess their contribution when combined with better-supported signals.
NEXT STEP	Focus Week 6 effort on the strongest observed factors and on validating decision-useful combinations.

3. INTERPRETATION GUARDRAILS**EVIDENCE LEVEL**

The results describe associations observed in the HealthConnect dataset. Correlation $\neq$ causality. The overall No-Show Rate of 51.15% is a descriptive reference, not a performance target. No recommendation introduces an unsupported numerical target.

DATA QUALITY

The appointment record remains the unit of analysis. Patient_id inconsistencies limit reliable reconstruction of patient-level histories; demographic and historical variables are therefore interpreted as recorded appointment-level attributes. Missing distance-to-clinic and waiting-time values were not imputed.

4. WEEK 5 DECISION DIRECTION**MONITOR → DIAGNOSE → PRIORITIZE**

MONITOR: quantify overall no-show exposure and reminder coverage. **DIAGNOSE:** identify factors that meaningfully differentiate No-Show Rates. **PRIORITIZE:** focus initial investigation and testing on longer booking lead times, using recorded previous no-shows as a supporting signal.

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